0620 • PAPER 4

TOPIC

01

States of matter

Questions only • Original examination pages retained

6
QUESTIONS

58
TOTAL MARKS

1
SECTIONS

USE THE TRACKER • ATTEMPT WITHOUT NOTES • MARK • REVIEW ERRORS

TOPIC CONTENTS

Quick navigation and progress overview

1.1 Particle model and changes of state

#	SOURCE	MARKS	DONE	MARK	REVIEW
01	2020 May/June Paper 42 Question 1	11	☐	☐	☐
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1 **(a)** Give the name of the process that:

(i) occurs when a gas turns into a liquid

.... [1]

(ii) occurs when a solid turns into a gas without first forming a liquid

.... [1]

(iii) is used to separate a mixture of liquids with different boiling points

.... [1]

(iv) is used to extract aluminium from aluminium oxide

.... [1]

(v) is used to separate a mixture of amino acids.

.... [1]

(b) The symbols of the elements in Period 2 of the Periodic Table are shown.

Li Be B C N O F Ne

For each of the following, give the symbol of an element from Period 2 which matches the description.

Each element may be used once, more than once or not at all.

Which element:

(i) combines with hydrogen to produce ammonia

.... [1]

(ii) makes up approximately 21% of clean, dry air

.... [1]

(iii) has atoms with only two electrons in the outer shell

.... [1]

(iv) has atoms with only seven protons

.... [1]

(v) is a monoatomic gas

.... [1]

(vi) is a soft metal stored in oil?

.... [1]

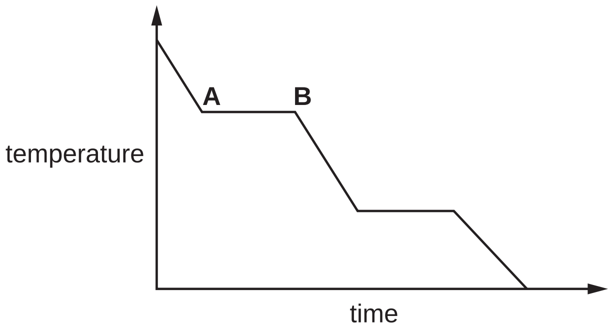
[Total: 11]

5 **(a)** Complete the table about solids, liquids and gases.

	particle separation	particle arrangement	type of motion
solid		regular	vibrate only
liquid	touching		random
gas	apart	random	

[3]

(b) The graph shows the change in temperature as a sample of a gas is cooled.



Name the change of state taking place between **A** and **B**.

.... [1]

(c) A bottle of liquid perfume is left open at the front of a room.

After some time, the perfume is smelt at the back of the room.

Name the **two** physical processes taking place.

1 ....

2 ....

[2]

[Total: 6]

1 This question is about states of matter.

(a) Complete the table, using ticks (✓) and crosses (✗), to describe the properties of gases, liquids and solids.

state of matter	particles are touching	particles have random movement	particles are regularly arranged
gas			
liquid			
solid			

[3]

(b) Substances can change state.

(i) Boiling and evaporation are two ways in which a liquid changes into a gas.

Describe **two** differences between boiling and evaporation.

1 ....

2 ....
[2]

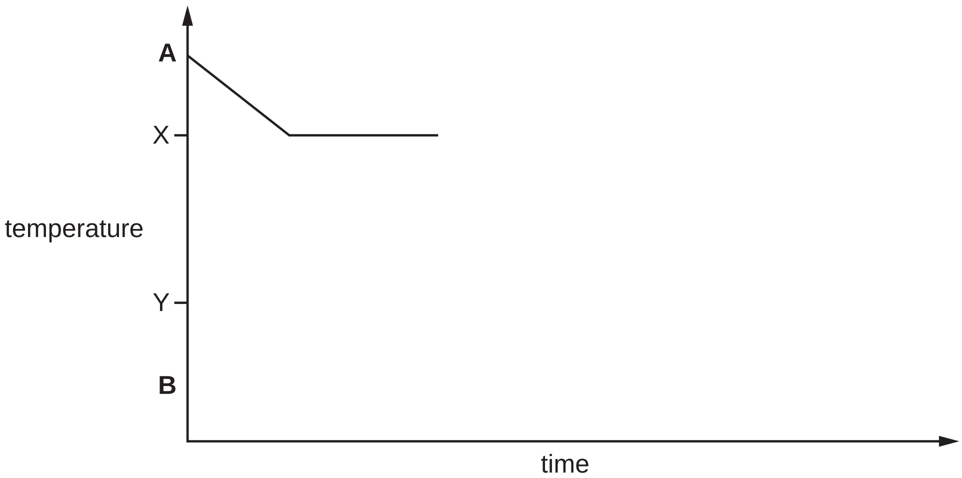
(ii) Name the change of state when:

- a gas becomes a liquid ....
- a solid becomes a gas. ....

[2]

(c) A substance boils at temperature X and melts at temperature Y.

Complete the graph to show the change in temperature over time as the substance cools from temperature **A** to temperature **B**.



[2]

(d) A solution is a mixture of a solute and a solvent.

(i) Name the process when a solid substance mixes with a solvent to form a solution.

.... [1]

(ii) Name the type of reaction when two solutions react to form an insoluble substance.

.... [1]

[Total: 11]

2 The table shows the melting points, boiling points and electrical conductivities of six substances, **D**, **E**, **F**, **G**, **H** and **I**.

substance	melting point / °C	boiling point / °C	conducts electricity when solid	conducts electricity when liquid
D	1083	2567	yes	yes
E	–117	79	no	no
F	3550	4827	no	no
G	119	445	no	no
H	–210	–196	no	no
I	801	1413	no	yes

(a) Identify the substance, **D**, **E**, **F**, **G**, **H** or **I**, which is:

(i) a liquid at 25 °C .... [1]

(ii) a gas at 25 °C .... [1]

(iii) a solid consisting of simple molecules at 25 °C. .... [1]

(b) Identify the substance, **D**, **E**, **F**, **G**, **H** or **I**, which is a metal. Give a reason for your choice.

substance ....

reason ....
[2]

(c) Identify the substance, **D**, **E**, **F**, **G**, **H** or **I**, which has a macromolecular structure. Give **two** reasons for your choice.

substance ....

reason 1 ....

reason 2 ....
[3]

(d) Identify the substance, **D**, **E**, **F**, **G**, **H** or **I**, which is an ionic solid. Give a reason for your choice.

substance ....

reason ....

....
[2]

[Total: 10]

1 Name the process used to:

(a) produce ammonia from nitrogen

.... [1]

(b) produce lead from molten lead(II) bromide

.... [1]

(c) separate an insoluble solid from a mixture of an insoluble solid and a solution

.... [1]

(d) produce ethanol from ethene

.... [1]

(e) identify the components of a mixture of soluble coloured substances

.... [1]

(f) separate a mixture of several liquids with different boiling points

.... [1]

(g) determine the volume of an acid required to neutralise a given volume of an alkali.

.... [1]

[Total: 7]

1 The three states of matter are solid, liquid and gas.

(a) Complete Table 1.1 to describe the structures of solids, liquids and gases in terms of particle separation, particle arrangement and particle motion.

Table 1.1

state of matter	particle separation	particle arrangement	particle motion
solid	touching		
liquid			
gas		random	random

[3]

(b) Substances can physically change.

Name each physical change:

- solid to liquid ....
- gas to liquid ....
- solid to aqueous. ....

[3]

(c) Gases diffuse.

(i) Describe, in terms of particles, why gases diffuse.

.... [1]

(ii) State what determines the relative rate of diffusion of gases at constant temperature.

.... [1]

(d) Gaseous oxides of nitrogen are atmospheric pollutants. One adverse effect of oxides of nitrogen is the formation of acid rain.

State **two other** adverse effects of oxides of nitrogen.

1..

2..

[2]

(e) In a catalytic converter, oxides of nitrogen are removed by reaction with the toxic gaseous product of the incomplete combustion of hydrocarbon fuels. Two non-toxic gases are formed in this reaction.

(i) Name the toxic gaseous product of the incomplete combustion of hydrocarbon fuels.

.... [1]

(ii) Name the **two** non-toxic gases formed in the catalytic converter.

1 ....

2 ....
[2]

[Total: 13]